



Division with Meaningful Remainders Teaching Bundle

Grade 4 • North Carolina • Mathematics

NC.4.NBT.6 • Explain whole-number division and remainders.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Explain whole-number division and remainders. This bundle practices a focused part of NC.4.NBT.6; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Build useful multiples of the divisor. Subtract partial groups. Check $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$.

Example 1: 79 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

$79 = 2 \times 39 + 1$; 1 is smaller than 2. Use 30 groups, then 9 more.

Example 2: 137 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

$137 = 2 \times 68 + 1$; 1 is smaller than 2. Use 60 groups, then 8 more.

Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

199 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 02

368 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 03

279 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 04

633 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.



Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

151 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 06

318 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 07

379 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 08

93 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.



Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

115 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 10

151 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 11

220 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 12

245 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.



Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

519 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 14

241 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 15

298 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.

CARD 16

29 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

Explain your method.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. 457 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 27 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. 109 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 307 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. 148 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 29 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. 128 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

8. 105 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. 53 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

10. 103 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. 352 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 175 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 450 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. 171 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

5. 47 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 186 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. 158 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 301 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 185 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 116 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Task-card worked solutions

1. 66 each; 1 left

$199 = 3 \times 66 + 1$; 1 is smaller than 3. Use 60 groups, then 6 more.

2. 61 each; 2 left

$368 = 6 \times 61 + 2$; 2 is smaller than 6. Use 60 groups, then 1 more.

3. 69 each; 3 left

$279 = 4 \times 69 + 3$; 3 is smaller than 4. Use 60 groups, then 9 more.

4. 79 each; 1 left

$633 = 8 \times 79 + 1$; 1 is smaller than 8. Use 70 groups, then 9 more.

5. 75 each; 1 left

$151 = 2 \times 75 + 1$; 1 is smaller than 2. Use 70 groups, then 5 more.

6. 63 each; 3 left

$318 = 5 \times 63 + 3$; 3 is smaller than 5. Use 60 groups, then 3 more.

Task-card worked solutions

7. 54 each; 1 left

$379 = 7 \times 54 + 1$; 1 is smaller than 7. Use 50 groups, then 4 more.

8. 46 each; 1 left

$93 = 2 \times 46 + 1$; 1 is smaller than 2. Use 40 groups, then 6 more.

9. 38 each; 1 left

$115 = 3 \times 38 + 1$; 1 is smaller than 3. Use 30 groups, then 8 more.

10. 75 each; 1 left

$151 = 2 \times 75 + 1$; 1 is smaller than 2. Use 70 groups, then 5 more.

11. 73 each; 1 left

$220 = 3 \times 73 + 1$; 1 is smaller than 3. Use 70 groups, then 3 more.

12. 61 each; 1 left

$245 = 4 \times 61 + 1$; 1 is smaller than 4. Use 60 groups, then 1 more.

Task-card worked solutions

13. 64 each; 7 left

$519 = 8 \times 64 + 7$; 7 is smaller than 8. Use 60 groups, then 4 more.

14. 48 each; 1 left

$241 = 5 \times 48 + 1$; 1 is smaller than 5. Use 40 groups, then 8 more.

15. 49 each; 4 left

$298 = 6 \times 49 + 4$; 4 is smaller than 6. Use 40 groups, then 9 more.

16. 14 each; 1 left

$29 = 2 \times 14 + 1$; 1 is smaller than 2. Use 10 groups, then 4 more.

Warm-up worked solutions

1. 76 each; 1 left

$457 = 6 \times 76 + 1$; 1 is smaller than 6. Use 70 groups, then 6 more.

2. 13 each; 1 left

$27 = 2 \times 13 + 1$; 1 is smaller than 2. Use 10 groups, then 3 more.

3. 27 each; 1 left

$109 = 4 \times 27 + 1$; 1 is smaller than 4. Use 20 groups, then 7 more.

4. 43 each; 6 left

$307 = 7 \times 43 + 6$; 6 is smaller than 7. Use 40 groups, then 3 more.

5. 18 each; 4 left

$148 = 8 \times 18 + 4$; 4 is smaller than 8. Use 10 groups, then 8 more.

6. 14 each; 1 left

$29 = 2 \times 14 + 1$; 1 is smaller than 2. Use 10 groups, then 4 more.

Warm-up worked solutions

7. 25 each; 3 left

$128 = 5 \times 25 + 3$; 3 is smaller than 5. Use 20 groups, then 5 more.

8. 17 each; 3 left

$105 = 6 \times 17 + 3$; 3 is smaller than 6. Use 10 groups, then 7 more.

9. 13 each; 1 left

$53 = 4 \times 13 + 1$; 1 is smaller than 4. Use 10 groups, then 3 more.

10. 17 each; 1 left

$103 = 6 \times 17 + 1$; 1 is smaller than 6. Use 10 groups, then 7 more.

Classwork worked solutions

1. 58 each; 4 left

$352 = 6 \times 58 + 4$; 4 is smaller than 6. Use 50 groups, then 8 more.

2. 58 each; 1 left

$175 = 3 \times 58 + 1$; 1 is smaller than 3. Use 50 groups, then 8 more.

3. 64 each; 2 left

$450 = 7 \times 64 + 2$; 2 is smaller than 7. Use 60 groups, then 4 more.

4. 24 each; 3 left

$171 = 7 \times 24 + 3$; 3 is smaller than 7. Use 20 groups, then 4 more.

5. 23 each; 1 left

$47 = 2 \times 23 + 1$; 1 is smaller than 2. Use 20 groups, then 3 more.

6. 46 each; 2 left

$186 = 4 \times 46 + 2$; 2 is smaller than 4. Use 40 groups, then 6 more.

Exit-ticket worked solutions

1. 52 each; 2 left

$158 = 3 \times 52 + 2$; 2 is smaller than 3. Use 50 groups, then 2 more.

2. 60 each; 1 left

$301 = 5 \times 60 + 1$; 1 is smaller than 5. Use 60 groups, then 0 more.

3. 30 each; 5 left

$185 = 6 \times 30 + 5$; 5 is smaller than 6. Use 30 groups, then 0 more.

4. 23 each; 1 left

$116 = 5 \times 23 + 1$; 1 is smaller than 5. Use 20 groups, then 3 more.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. 116 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 539 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 63 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 362 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. 136 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 284 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

7. 383 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

8. 397 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Fall • Worked solutions

1. 14 each; 4 left

$116 = 8 \times 14 + 4$; 4 is smaller than 8. Use 10 groups, then 4 more.

2. 67 each; 3 left

$539 = 8 \times 67 + 3$; 3 is smaller than 8. Use 60 groups, then 7 more.

3. 31 each; 1 left

$63 = 2 \times 31 + 1$; 1 is smaller than 2. Use 30 groups, then 1 more.

4. 60 each; 2 left

$362 = 6 \times 60 + 2$; 2 is smaller than 6. Use 60 groups, then 0 more.

Fall • Worked solutions

5. 45 each; 1 left

$136 = 3 \times 45 + 1$; 1 is smaller than 3. Use 40 groups, then 5 more.

6. 47 each; 2 left

$284 = 6 \times 47 + 2$; 2 is smaller than 6. Use 40 groups, then 7 more.

7. 76 each; 3 left

$383 = 5 \times 76 + 3$; 3 is smaller than 5. Use 70 groups, then 6 more.

8. 56 each; 5 left

$397 = 7 \times 56 + 5$; 5 is smaller than 7. Use 50 groups, then 6 more.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. 83 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 313 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 53 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 87 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. 254 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 89 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

7. 220 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

8. 127 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Winter • Worked solutions

1. 16 each; 3 left

$83 = 5 \times 16 + 3$; 3 is smaller than 5. Use 10 groups, then 6 more.

2. 52 each; 1 left

$313 = 6 \times 52 + 1$; 1 is smaller than 6. Use 50 groups, then 2 more.

3. 26 each; 1 left

$53 = 2 \times 26 + 1$; 1 is smaller than 2. Use 20 groups, then 6 more.

4. 17 each; 2 left

$87 = 5 \times 17 + 2$; 2 is smaller than 5. Use 10 groups, then 7 more.



Winter • Worked solutions

5. 63 each; 2 left

$254 = 4 \times 63 + 2$; 2 is smaller than 4. Use 60 groups, then 3 more.

6. 44 each; 1 left

$89 = 2 \times 44 + 1$; 1 is smaller than 2. Use 40 groups, then 4 more.

7. 36 each; 4 left

$220 = 6 \times 36 + 4$; 4 is smaller than 6. Use 30 groups, then 6 more.

8. 15 each; 7 left

$127 = 8 \times 15 + 7$; 7 is smaller than 8. Use 10 groups, then 5 more.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. 152 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 350 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 399 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 123 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. 272 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 299 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

7. 61 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

8. 629 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Spring break • Worked solutions

1. 30 each; 2 left

$152 = 5 \times 30 + 2$; 2 is smaller than 5. Use 30 groups, then 0 more.

2. 43 each; 6 left

$350 = 8 \times 43 + 6$; 6 is smaller than 8. Use 40 groups, then 3 more.

3. 66 each; 3 left

$399 = 6 \times 66 + 3$; 3 is smaller than 6. Use 60 groups, then 6 more.

4. 61 each; 1 left

$123 = 2 \times 61 + 1$; 1 is smaller than 2. Use 60 groups, then 1 more.

Spring break • Worked solutions

5. 45 each; 2 left

$272 = 6 \times 45 + 2$; 2 is smaller than 6. Use 40 groups, then 5 more.

6. 59 each; 4 left

$299 = 5 \times 59 + 4$; 4 is smaller than 5. Use 50 groups, then 9 more.

7. 20 each; 1 left

$61 = 3 \times 20 + 1$; 1 is smaller than 3. Use 20 groups, then 0 more.

8. 78 each; 5 left

$629 = 8 \times 78 + 5$; 5 is smaller than 8. Use 70 groups, then 8 more.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. 190 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 178 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 98 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 547 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. 73 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

6. 51 counters are shared equally among 2 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

7. 177 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

8. 261 counters are shared equally among 7 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Summer • Worked solutions

1. 63 each; 1 left

$190 = 3 \times 63 + 1$; 1 is smaller than 3. Use 60 groups, then 3 more.

2. 44 each; 2 left

$178 = 4 \times 44 + 2$; 2 is smaller than 4. Use 40 groups, then 4 more.

3. 16 each; 2 left

$98 = 6 \times 16 + 2$; 2 is smaller than 6. Use 10 groups, then 6 more.

4. 78 each; 1 left

$547 = 7 \times 78 + 1$; 1 is smaller than 7. Use 70 groups, then 8 more.

Summer • Worked solutions

5. 36 each; 1 left

$73 = 2 \times 36 + 1$; 1 is smaller than 2. Use 30 groups, then 6 more.

6. 25 each; 1 left

$51 = 2 \times 25 + 1$; 1 is smaller than 2. Use 20 groups, then 5 more.

7. 44 each; 1 left

$177 = 4 \times 44 + 1$; 1 is smaller than 4. Use 40 groups, then 4 more.

8. 37 each; 2 left

$261 = 7 \times 37 + 2$; 2 is smaller than 7. Use 30 groups, then 7 more.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Explain whole-number division and remainders. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Build useful multiples of the divisor. Subtract partial groups. Check $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. 248 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 427 counters are shared equally among 6 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 37 counters are shared equally among 3 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 92 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Substitute lesson • Worked solutions

1. 41 each; 2 left

$248 = 6 \times 41 + 2$; 2 is smaller than 6. Use 40 groups, then 1 more.

2. 71 each; 1 left

$427 = 6 \times 71 + 1$; 1 is smaller than 6. Use 70 groups, then 1 more.

3. 12 each; 1 left

$37 = 3 \times 12 + 1$; 1 is smaller than 3. Use 10 groups, then 2 more.

4. 18 each; 2 left

$92 = 5 \times 18 + 2$; 2 is smaller than 5. Use 10 groups, then 8 more.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Explain whole-number division and remainders. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Build useful multiples of the divisor. Subtract partial groups. Check $\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}$.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. 118 counters are shared equally among 8 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

2. 238 counters are shared equally among 5 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

3. 134 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.

4. 310 counters are shared equally among 4 groups. Each group must get a whole number. How many per group and how many left? Show a partial-quotient strategy.



Emergency substitute lesson • Worked solutions

1. 14 each; 6 left

$118 = 8 \times 14 + 6$; 6 is smaller than 8. Use 10 groups, then 4 more.

2. 47 each; 3 left

$238 = 5 \times 47 + 3$; 3 is smaller than 5. Use 40 groups, then 7 more.

3. 33 each; 2 left

$134 = 4 \times 33 + 2$; 2 is smaller than 4. Use 30 groups, then 3 more.

4. 77 each; 2 left

$310 = 4 \times 77 + 2$; 2 is smaller than 4. Use 70 groups, then 7 more.

A completed lesson-plan example

Level and focus

Grade 4; NC.4.NBT.6; Explain whole-number division and remainders.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.4.NBT.6 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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